

CLIENT REQUIREMENTS DOCUMENT

Electronic Records And Document Management System (ERDMS)

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I. Document Information

Project Title:	Electronic Records and Document Management System (ERDMS)
System Type:	Web-Based Full Stack Application
Technology Stack:	Angular + Node.js + Express.js + MySQL
Document Type:	Client Requirements Specification
Prepared For:	CPE 8 Software Design
Development Model:	Software Development Life Cycle (SDLC) - Waterfall Methodology

II. Project Overview

The **Electronic Records and Document Management System (ERDMS)** is a **centralized, secure, and role-based web application** designed to efficiently manage organizational documents and records in a digital environment.

This system replaces traditional **manual and paper-based document handling**, reducing inefficiency, duplication, and data loss. It provides a structured platform where users can:

- Upload documents
- Store files securely
- Retrieve documents easily
- Manage and organize records systematically

The system is designed with **role-based access control**, ensuring that only authorized users can access specific functionalities.

User Roles:

- **Administrator (Admin)** – Full system control and management
- **Standard User (User)** – Limited access for document handling

Each role is assigned specific permissions to ensure **data security, integrity, and proper system governance**.

III. Project Objectives

The ERDMS aims to achieve the following objectives:

- Develop a centralized and structured digital document management system
- Implement secure role-based access control for system users

- Enable efficient uploading, storage, and retrieval of documents
- Provide real-time access to documents for authorized users
- Track system activities through logs and audit trails
- Improve organizational efficiency in document handling and retrieval
- Apply full stack development principles in a real-world application
- Demonstrate integration of frontend, backend, and database systems

IV. System Scope

The ERDMS includes the following core modules and functionalities:

User Management

This module handles all user-related operations in the system.

- User registration and authentication (login/logout)
- Role assignment (Admin / User)
- Account management (update profile, status control)

Document Management

This is the core functionality of the system, responsible for handling all document operations.

- Uploading documents securely into the system
- Viewing and accessing uploaded files
- Downloading documents when needed
- Deleting documents (Admin-only permission)
- Categorizing documents for better organization

System Monitoring

This module ensures system transparency and accountability.

- Activity logging for all user actions
- Audit trail tracking (who accessed or modified data)
- Login and logout tracking history

Dashboard

Provides a visual summary of system activity and statistics.

- Total number of users
- Total uploaded documents
- Recent system activities
- Quick access navigation for admin functions

V. Stakeholders

Stakeholders are individuals or groups who have an interest in the development, implementation, and use of the system. Each stakeholder plays a specific role in ensuring the success of the project.

Stakeholder	Role and Responsibilities
System Administrator	Responsible for overall system management, including user account control, document management, system configuration, and monitoring system security and activity logs.
Registered Users	End-users of the system who can access, view, upload (if permitted), and download documents based on assigned permissions.
Instructor	Evaluates the system based on requirements, functionality, code quality, and documentation. Provides guidance and grading for the project output.
Students (Developers)	Responsible for designing, developing, testing, and deploying the system using full stack technologies, ensuring all requirements are properly implemented.

VI. User Roles and Permissions

The system follows a **Role-Based Access Control (RBAC)** approach, where users are assigned specific roles with defined permissions. This ensures system security, data integrity, and controlled access to functionalities.

Admin Role

The **Administrator (Admin)** is the highest-level user in the system. This role has **full control over all system functionalities**, including user management, document management, and system monitoring.

Functional Requirements:

1. User Management

The Admin is responsible for managing all user accounts within the system.

- Create new user accounts
- Update user details (name, role, status)
- Delete user accounts

- Assign roles (Admin or User)
- Activate or deactivate user accounts

This ensures proper control over who can access the system.

2. Folder Management

The Admin can organize the system structure through folder management to improve document organization and accessibility.

- Create new folders for document categorization
- Rename existing folders
- Delete folders (including contained documents if allowed)
- Assign documents to specific folders
- Manage folder hierarchy (if applicable)
- Control folder access permissions

This feature ensures that documents are **systematically organized and easy to locate**, especially in large-scale systems.

3. Document Management

The Admin has full authority over all system documents.

- Upload new documents into the system
- Edit document metadata (title, category, description)
- Delete documents from the system
- Assign or update document categories
- Control document accessibility (who can view/download)

This ensures proper organization and security of all stored files.

4. Dashboard Access

The Admin has access to system-wide analytics and overview information.

- View total number of registered users
- View total number of uploaded documents
- View recent system activities
- View system usage statistics

This provides insight into overall system performance and usage.

5. Logs and Audit Trail

The Admin can monitor all system activities for security and tracking purposes.

- View user login and logout history
- Track document uploads and downloads
- Monitor user activities within the system
- View system access logs

This ensures accountability and system transparency.

User Role

The **Standard User** is an end-user with limited access to system functionalities. This role focuses mainly on **document access and personal account management**.

Functional Requirements:

1. User Management

Users can manage their own personal account information.

- View personal profile
- Update personal information (e.g., name, email)
- Change password for security

2. Document Management

Users are allowed to interact with documents based on permissions granted by the Admin.

- View available documents
- Download permitted documents
- Search documents by keyword
- Filter documents by category

Users can only access documents; they cannot modify system data.

3. Restrictions (User Limitations)

To maintain system security and proper control, standard users are NOT allowed to:

- Upload documents
- Delete documents
- Manage user accounts
- View system logs or audit trails
- Access admin dashboard

VII. Non-Functional Requirements

Non-functional requirements define the quality and performance standards of the system.

Performance

- System must respond within 2–3 seconds
- Must support multiple concurrent users
- Efficient database queries for fast data retrieval

Security

- Passwords must be encrypted using secure hashing algorithms

- Role-based access control must be strictly enforced
- All API endpoints must be protected and validated

Usability

- System must have a simple and user-friendly interface
- Responsive design for both desktop and mobile devices
- Intuitive navigation for all user roles

Maintainability

- System must follow a modular code structure
- Code should be easy to update and extend
- Proper separation of frontend, backend, and database logic

Scalability

- System must support future feature expansion
- Should handle increased number of users and documents
- Architecture must allow integration of new modules

VIII. System Architecture

The system follows a standard **three-tier architecture model** commonly used in full stack development.

```
Frontend (Angular)
    ↓
Backend (Node.js + Express API)
    ↓
Database (MySQL)
    ↓
File Storage System (Local Server / Upload Directory)
```

Frontend Layer

- Built using Angular
- Handles user interface and interaction
- Sends requests to backend APIs

Backend Layer

- Built using Node.js + Express.js
- Processes business logic and API requests
- Communicates with database and file system

Database Layer

- Uses MySQL
- Stores structured data such as users and documents
- Ensures data integrity and relationships

File Storage Layer

- Stores uploaded documents in a local or server directory
- Linked with database through file references

*****End of document*****